# Student Academic Performance & Learning Risk Analysis

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**Submission : 19th October 2026**

# INTRODUCTION

Several factors such as study habits, attendance, assignment scores, and internet usage influence a student's academic performance. We can identify patterns in learning behaviours and recognize students who may benefit from additional academic support by understanding how these factors relate to students' academic performances.

This project, Student Academic Performance and Learning Risk Analysis, aims to explore the relationships between a student's academic and behavioral characteristics by analysing the data and using Machine Learning techniques. The dataset provided contains 200 students, containing their data such as age, gender, weekly study hours, attendance percentage, assignment scores, previous exam scores, sleep hours, internet usage hours, and academic risk categories.

We are using Exploratory Data Analysis (EDA) to identify trends and patterns in the students' dataset, K-Means clustering to group students with similar characteristics, and Decision Tree Classification to examine predict each student's academic-risk category as Low, High or Medium. This project's objective is to develop a better understanding of student learning patterns and highlight factors affecting academic performances and provide insights on how data analysis could contribute in academic decision-making and strategies.

## Descriptive Statistical Analysis

We now calculated descriptive statistics for the numerical variables in the given dataset. The measures examined were mean, median, minimum, maximum, and sample standard deviation.

These statistics provide an overview of the student's learning patterns, ranges and variability of the student characteristics in academics. These results will help identify patterns and EDA.

## Results and interpretation:

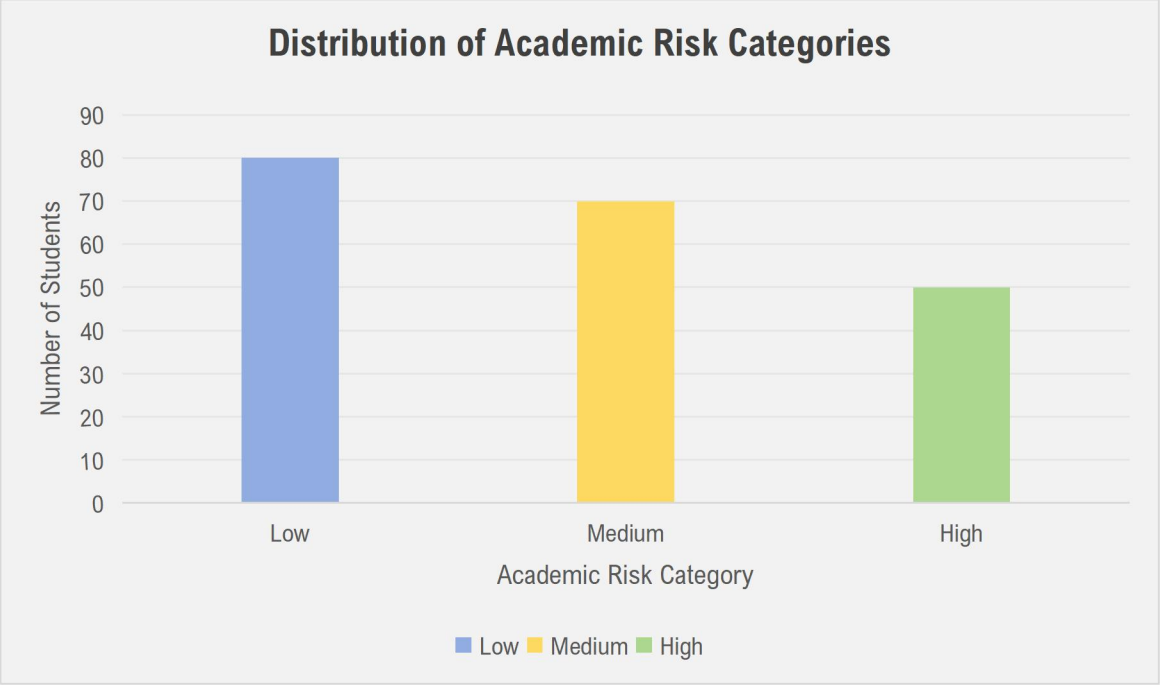
| Variable | Mean | Median | Minimum | Maximum | Standard Deviation |
|---|---|---|---|---|---|
| Age | 19.75 | 20 | 17 | 22 | 1.761537702 |
| StudyHoursPerWeek | 11.7505 | 11.45 | 1 | 23.5 | 5.31799824 |
| AttendancePercentage | 77.101 | 79.75 | 40 | 100 | 14.42746191 |
| AssignmentScore | 70.745 | 73.25 | 33.2 | 100 | 16.24385388 |
| PreviousExamScore | 67.0375 | 69.6 | 28.2 | 97.2 | 16.34672508 |
| SleepHours | 6.869 | 7 | 4 | 8.8 | 0.982939392 |
| InternetUsageHours | 4.2615 | 4.2 | 0.5 | 9.8 | 1.937715525 |

# EXPLORATORY DATA ANALYSIS(EDA):

We analysed the distribution of academic risk by counting the students classified as Low, Medium and High risk. The frequency and percentage of each category were calculated and visualized by using a bar chart.

| Academic Risk | Student Count | Percentage(%) |
|---|---|---|
| Low | 80 | 40 |
| Medium | 70 | 35 |
| High | 50 | 25 |

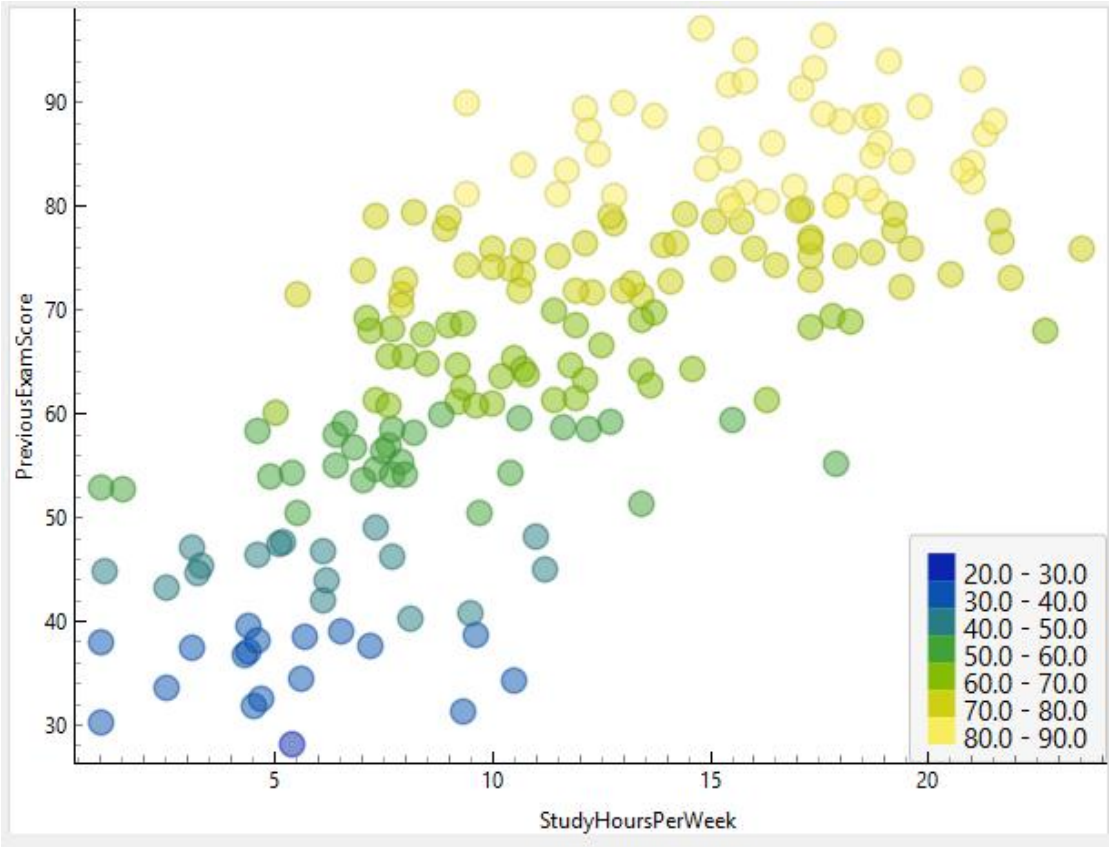
The analysis identified 80 low-risk students (40%), 70 medium-risk students (35%), and 50 high-risk students (25%).




The distribution provides an overview of the representation of each risk category in the dataset and helps determine whether the categories are balanced or imbalanced. When interpreting the results of the subsequent Decision tree classification model, this information is relevant.

## Relationship Between Study Hours and Previous Exam Scores

To examine the relationship between students' weekly study hours and their previous exam scores, we created a scatter plot. Each point represents an individual student.

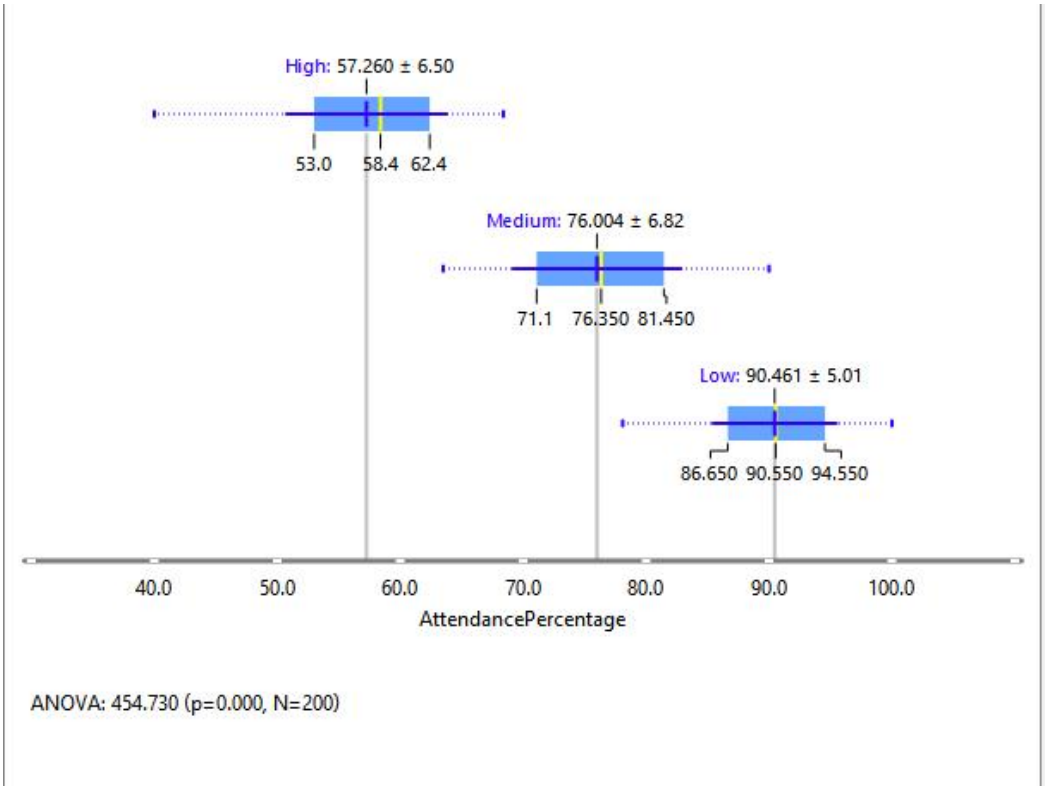



The plot shows a positive relationship : students with higher weekly study hours tend to have higher previous examination scores. The visual pattern is consistent with the previously calculated Pearson correlation coefficient of 0.7283.. It indicates a strong positive linear association in this dataset.

However, one might notice that the data points do not lie on a perfect straight line. This implies that the marks obtained by the students are not solely based on the study hours they put in. The study hours cannot completely account for the variability in academic performance.

## Relationship between Attendance and Academic Risk:

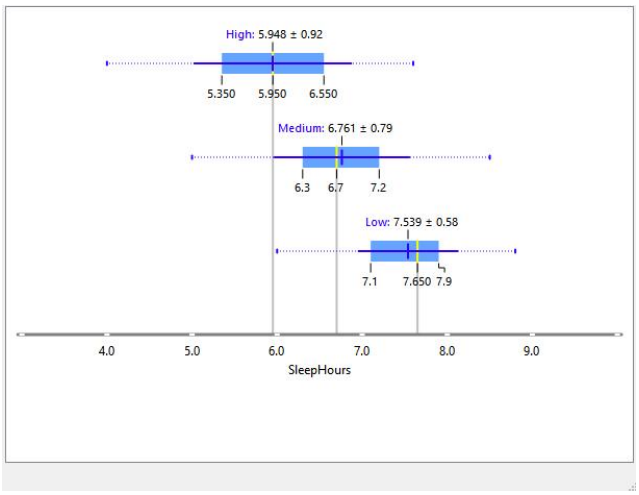
Box plots were employed to compare the attendance percentages of the students who are High, Medium, and Low academic-risk levels. The attendance median for the High risk level was 58.4%, while that of the Medium risk level was 76.35% and Low risk level was 90.55%.




The findings clearly indicate that there is an association between poor attendance and academic risk among students. In the boxplot provided, it is clear that the Low-risk group had the highest median value for attendance, whereas the High-risk group had the lowest. This indicates that attendance might be a vital variable related to academic risk among students. Nonetheless, it is crucial to understand that the correlation established here does not imply causality.

## Relationship Between Sleep Hours and Academic Risk:

Now, we use a box plot to compare the sleep duration of the students across the High, Medium and Low academic-risk groups. The median sleep duration was 5.95 hours for the high-risk group, 6.70 hours for the medium -risk groups and 7.65 hours for the low-risk groups.



The results show that the High-risk group had the lowest median sleep duration, while the Low-risk group had the highest. Moreover, the middle 50% of the distribution of values of sleep durations was different for the three groups as well. However, the analysis does not establish that sleep duration directly causes differences in academic performance.

## Relationship Between Internet Usage and Academic Risk

We now use a box plot to compare internet usage hours across the High, Medium, and Low academic-risk categories. The reported median internet usage was 6.35 hours for the High-risk group, 4.50 hours for the Medium-risk group, and 2.60 hours for the Low-risk group.

The findings indicate that there is a declining trend of internet use from the High-risk group to the Low-risk group. The high number of internet users among students in the High-risk group indicates that those in the High-risk group were using the internet the most compared to those in the Low-risk group, who used the internet the least. There is an indication of correlation between internet use and academic risk.

# METHODOLOGY:

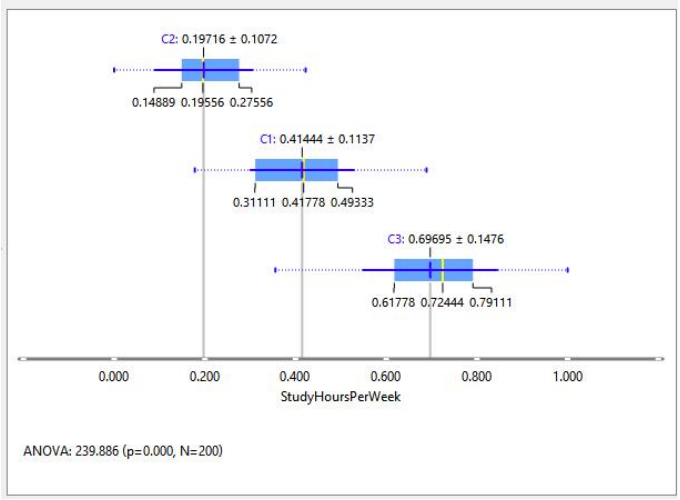
## A) K-Means Clustering :

K-Means clustering was performed using Orange Data Mining. StudentID, Gender, and Age were excluded, while six numerical study - related and lifestyle variables were selected as clustering features. The numerical features were normalized to make their scales comparable, and AcademicRisk was retained separately for interpretation. The number of clusters was set to three.

Orange started off with initialization of the cluster centers and then repeatedly assigned the students to clusters depending on their Euclidean distance from each cluster center till the algorithm's stoppage criteria were fulfilled. These clusters were then analyzed using boxplots in order to get an idea about their nature.

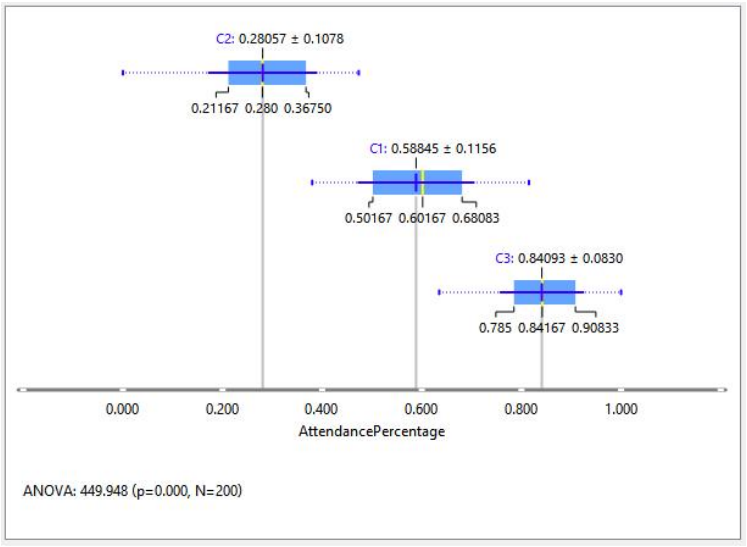
## Study hours

In terms of normalized study hours for each cluster, C2 was found to have the minimum value (0.1972), then C1 (0.4144), whereas the maximum value was found for C3 (0.6970). This shows that there is a difference in the pattern of study hours among the three clusters. More comparisons need to be made in order to define the cluster characteristics.



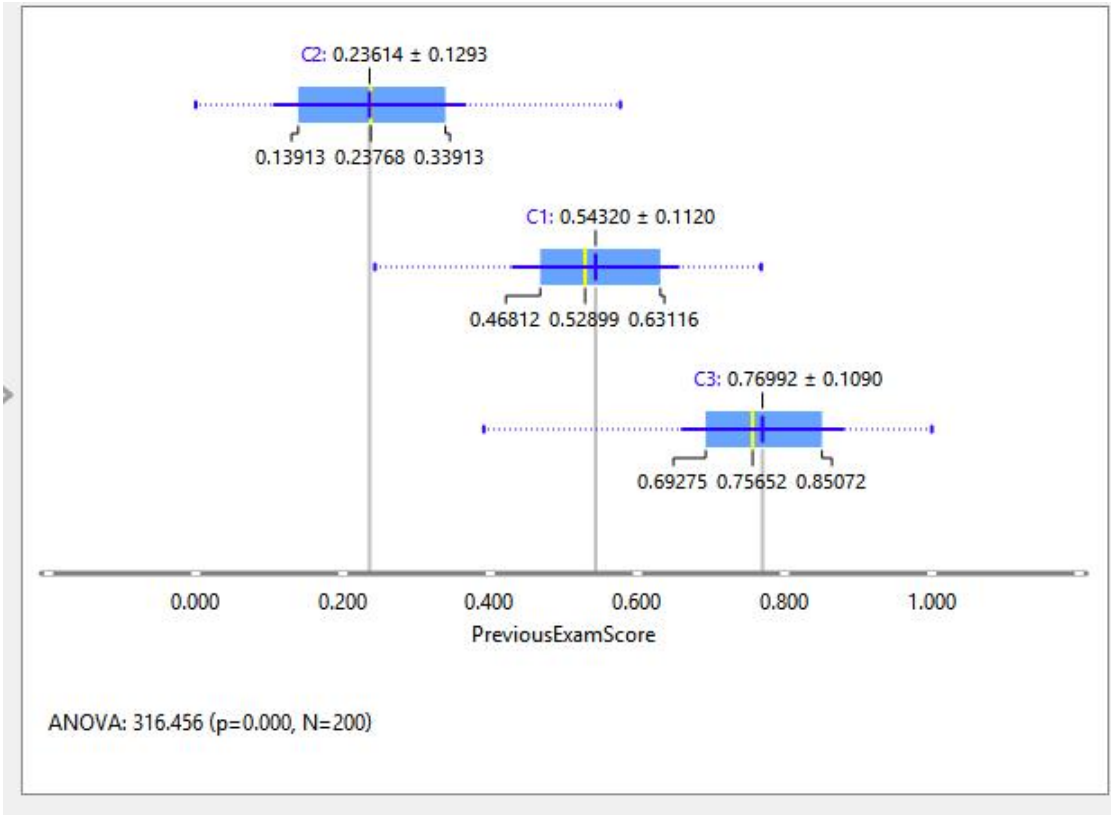

## Attendance Percentage:

Comparison between normalized values of attendance showed significant differences. Cluster C2 was found to have the smallest normalized value (0.2806) followed by cluster C1 (0.5885), while cluster C3 had the highest normalized value (0.8409). In conjunction with the results obtained on study-hour variable, this means that C3 is characterized by students having high study hour and attendance values compared to the other two clusters, while C2 has relatively low values of the variables. Academic performance and lifestyle analysis will be conducted to further classify the clusters.



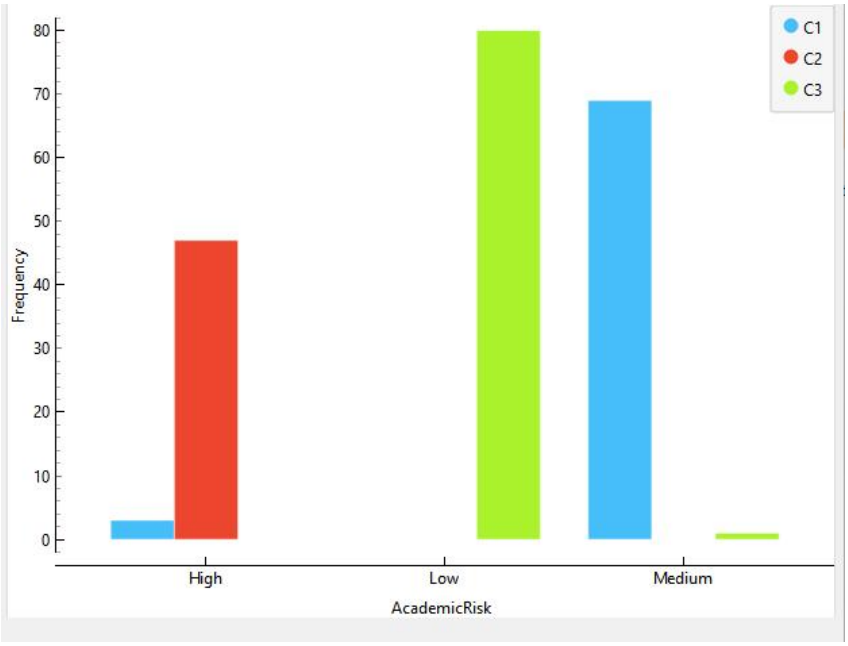

## Previous Exam score:

The comparison of normalized previous exam scores showed differences among the three clusters. C2 had the lowest reported value (0.2361), followed by C1 (0.5432), while C3 had the highest value (0.7699). This follows the same pattern observed for study hours and attendance. C3 has relatively higher values across all three variables, whereas C2 has relatively lower values. These findings suggest that the clustering has identified groups with different academic-behaviour patterns.



## Comparison with Academic Risk

The three clusters were compared with the original AcademicRisk categories to interpret the groups. C1 contained 69 Medium-risk and 3 High-risk students. C2 contained all 47 students assigned to that cluster in the High-risk category, while C3 contained all 81 Low-risk students. These results show a strong association between the clusters and the original academic-risk categories. The clusters were developed based on variables related to the research as well as the participant's lifestyle but without the inclusion of AcademicRisk in the process of clustering because it was only used for comparison purposes. Nonetheless, the relationship established is not a causal one.




## B) Decision Tree Classification:

The input features were StudyHoursPerWeek, AttendancePercentage, AssignmentScore, PreviousExamScore, SleepHours, InternetUsageHours, and Gender. StudentID and Age were not included. AcademicRisk was chosen as the output. Gender was numerically encoded using categorical preprocessing.

## Evaluation Results:

- The model correctly classified approximately 77% of the student records.
- F1 score being 0.754, this balances precision and recall, which gives an overall measure of classification.
- Precision is 0.771. The reported precision indicates how often predictions were correct when the model assigned a class, average according to Orange's scoring method

- Recall is reported to be 0.770. It measures how clearly it identified students belonging to their actual risk categories, averaged according to Orange's scoring method.

Evaluation results for target (None, show average over classes)

| Model | AUC | CA | F1 | Prec | Recall | MCC |
|---|---|---|---|---|---|---|
| Tree (1) | 0.835 | 0.770 | 0.754 | 0.771 | 0.770 | 0.662 |

```
How does the average InternetUsageHours differ between High and
Low AcademicRisk students?
Use arrowsandto navigate
```

```
Options
How does the average InternetUsageHours differ between High and
Low AcademicRisk students?
```

## Confusion Matrix:

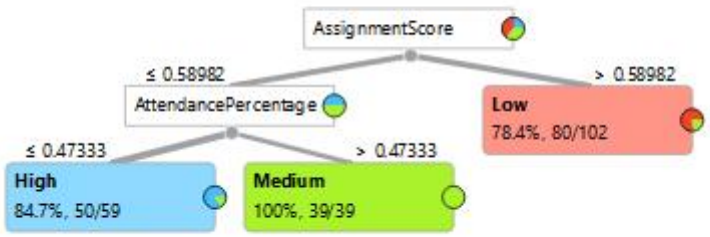
Predicted

| Actual | High | Low | Medium | Σ |
|---|---|---|---|---|
| High | 42 | 0 | 8 | 50 |
| Low | 0 | 78 | 2 | 80 |
| Medium | 11 | 25 | 34 | 70 |
| Σ | 53 | 103 | 44 | 200 |

Select Correct | Select Misclassified | Clear Selection

- The model identified 42 out of 50 High-Risk students. It misclassified the other 8 as Medium Risk.
- The model identified 78 of 80 Low-risk students. Only 2 were classified as Medium risk.
- The model identified 34 of 70 Medium-risk students. It classified 11 as High risk and 25 as Low risk.

Main finding : The model separates Low-risk students very well, identifies most High-risk students, but struggles to separate Medium-risk students from the other two categories.




- Assignment Score was identified as the Root node in the decision tree. This indicates that it was selected for the first splitStudents with a normalised AssignmentScore greater than 0.58982 were classified as Low Academic Risk, with 78.4% of the 102 students in this leaf belonging to this group.
- The tree used Attendance Percentage for a further split for students with an Assignment score less than or equal to 0.58982. Students were classified as High academic risk for students whose normalised Attendance percentage was less than or equal to 0.47333. The leaf contained 50 out of 59 students(84.7%).
- Students with an AttendancePercentage greater than 0.47333 were classified as Medium Academic Risk, and all 39 students in that leaf belonging to the Medium category.
- These rules show how the model combines assignment performance and attendance to classify academic risk.

# RESULTS:

## A) K-Means Clustering Results

The K-Means algorithm was configured to form three clusters using six numerical features from the dataset: StudyHoursPerWeek, AttendancePercentage, AssignmentScore, PreviousExamScore, SleepHours, and InternetUsageHours. The features were normalized before clustering, and StudentID, Age, and Gender were excluded from the clustering features using Orange. AcademicRisk was not used to create the clusters; it was used afterward to interpret them.

The algorithm showed the following clusters:

- Cluster 1: 72 students (36%)
- Cluster 2: 47 students (23.5%)
- Cluster 3: 81 students (40.5%)

Cluster 1 - Moderate-risk group: This cluster contained 72 students. It showed intermediate values for study hours, attendance, and previous exam scores of the students. From these students, 69 were labelled Medium risk and 3 were labelled High risk in the original dataset.

Cluster 2 - High-risk group: This cluster showed 47 students and showed the lowest values for study hours, attendance, and previous exam scores among the three clusters. All 47 students were labelled High risk in the original dataset.

Cluster 3 - low-risk group: This cluster contained 81 students and showed the highest values for study hours, attendance, and previous exam scores. Of these students, 80 were labelled Low risk and 1 was labelled Medium risk.

The cluster results showed a strong association between the groups formed by K-Means and the existing AcademicRisk labels. C2 correlate to the High-risk group, C1 largely corresponded to the Medium-risk group, and C3 largely corresponded to the Low-risk group. These findings describe patterns in this dataset and do not establish that any individual factor causes academic risk.

## B) Decision Tree Classification Results:

The Decision Tree selected AssignmentScore as its root node,and it is followed by AttendancePercentage as the next node. This shows that these features were important in the splits formed by this particular tree. The model used six numerical

features and the encoded Gender feature to classify students into High, Medium, and Low AcademicRisk categories.

## Example Decision Paths and Prediction Rules

The Tree Viewer displayed the following decision paths:

- If AssignmentScore is greater than 0.58982, predict Low Academic Risk.
- If Assignment Score ≤ 0.58982 and Attendance Percentage ≤ 0.47333, predict High Academic Risk.
- If Assignment Score ≤ 0.58982 and Attendance Percentage > 0.47333, predict Medium Academic Risk

These cut-off values are based on the normalized feature range displayed by Orange and not on the actual assignment score or attendance percentage values.

## Model Performance:

| Evaluation Metric | Results |
|---|---|
| Classification Accuracy (CA) | 77% |
| F1 score | 0.754 |
| Precision | 0.771 |
| Recall | 0.770 |

The confusion matrix showed that the model accurately classified 42 out of 50 High-risk students, 78 out of 80 Low-risk students, and 34 out of 70 Medium-risk students. The model classified Low-risk students most successfully, while Medium-risk students were the most challenging category to distinguish.

## Overall Interpretation

From the Decision Tree analysis, it can be seen that both assignment performance and attendance are good predictors to differentiate academic-risk categories from this data set. However, the Decision Tree was not able to classify all students accurately, especially the students categorized as medium risk. It means that the results obtained need to be viewed as exploratory classifications.

# INSIGHTS AND LEARNINGS

## Factors associated with academic risk:

Exploratory Data Analysis (EDA) revealed clear associations between academic risk and several student - related factors. Students in the High-risk group recorded relatively low median attendance rates of about 58.40 % and low median sleeping time of about 5.95 hours as compared to the Low-risk group, which had a median attendance rate of about 90.5% and a median sleeping time of about 7.65 hours. In addition, the use of the internet among the two categories was not the same since the median for the High-risk group was 6.35 while that of the Low-risk category was 2.6. This shows that attendance, sleep, internet use, study time, and academic performance are crucial determinants of academic risk.

## How did clustering reveal hidden groups?

K-K-Means clustering grouped students into 3 clusters based on similarities across six numerical features, without using the AcademicRisk label(target) to form the clusters. After clustering, comparing the groups with the original labels revealed that one cluster corresponded predominantly to Medium-risk students, another consisted entirely of High-risk students, and the third corresponded almost entirely to Low-risk students. This demonstrated how unsupervised learning can reveal patterns in student data and identify groups with similar characteristics using the dataset.

## How does a Decision Tree make its predictions understandable?

The Decision Tree made its predictions understandable by dividing students into groups through a series of feature - based rules. In this model, AssignmentScore was selected for the first split(root node ), followed by AttendancePercentage. The resulting branches led to predictions of High, Medium, or Low AcademicRisk.In the case of using a tree structure from the root node to the leaf node, it became possible to analyze the reasoning behind the prediction made for a particular student. The algorithm was 77% accurate in classifying students; however, the medium-risk group was harder to classify.

## What did you learn about preprocessing, distance, and impurity?

Preprocessing was efficient because the numerical features had different scales. Normalization brought the selected features onto a common 0–1 scale, helping prevent features with larger numerical ranges from dominating distance calculations in K-Means. K-Means used similarities between students, represented through distances in the feature space to form clusters. The choice of features and scaling can therefore also influence the resulting groups.

The Decision Tree used feature-based splits to separate students into groups with more consistent AcademicRisk labels. The principle of impurity explains how a useful split produces child groups that are more homogeneous than the parent group. Measures such as Gini impurity and entropy are commonly used to explain this concept.

This project showed the possibility of combining EDA, unsupervised clustering, and supervised classification to explore patterns of academic risks of the students. The necessity of choosing suitable features, preparing data, evaluating the models, and making cautious conclusions was emphasized as well. Models based on machine learning may reveal some patterns in the data, but their predictions shouldn't be considered definite evaluations of students.

# CHALLENGES AND RECOMMENDATIONS

## Challenges:

### 1. Distance calculations in K-Means:

K - Means grouped students based on distances between their feature values. Understanding how these distances determine cluster membership was challenging, particularly when multiple numerical features were considered simultaneously. Orange Data Mining simplified the process by performing the calculations automatically.

### 2. Feature Encoding and Normalization

The dataset contained numerical and categorical features. Gender required encoding before it could be included in the Decision Tree model, while numerical features were normalized before K-Means clustering. These preprocessing steps are important because features with different numerical scales can influence some machine-learning algorithms differently [1].

### 3. Choosing a Suitable Value of K

Choosing the number of clusters was a challenge in K-Means. We chose three clusters to investigate whether the students could be grouped into High-, Medium-, and Low-risk patterns. The silhouette method can help evaluate how well-separated clusters are and compare different choices of K [2].

### 4. Understanding Impurity in Decision Trees

Understanding how a Decision Tree selects feature-based splits required learning about class purity and impurity measures such as Gini impurity and entropy. The working of the model assigning predicted academic risk categories was explained by interpreting the branches and leaf nodes. IBM explains that *Gini impurity* and information gain are commonly used to assess the quality of potential Decision Tree splits [6].

## Recommendations

### 1. Use a Larger Dataset in Future Studies:

Further investigations need to involve more students from various classes and different schools. Such a wider database might be used to assess whether the same patterns are valid for other students besides the currently involved 200. Training data should represent the population to which a machine-learning model will be applied, helping reduce potential bias [5].

## 2. Include Additional Relevant Features

Other factors that might give us a better picture of the academic experience of the students include their participation in class, revision techniques, recalling abilities, homework tendencies, and involvement in extra-curricular activities.

## 3. Compare Different Machine-Learning Algorithms

For future studies, comparisons between K-Means and other algorithms for clustering can be made. Also, comparisons can be made between the Decision Tree and algorithms like Random Forest, Logistic Regression, and k-Nearest Neighbors.

## 4. Evaluate the Choice of K Systematically

Future experiments should compare multiple values of K using methods such as the elbow method and silhouette analysis. The silhouette method evaluates how closely each sample fits its assigned cluster compared with neighbouring clusters [2]. This will yield better evidence regarding the choice of appropriate clusters.

## 5. Use Predictions Responsibly

The prediction made by the model should not be considered the only thing upon which critical academic decisions will be based because the predictions can be wrong, especially those that relate to students classified as Medium risk. It is important to exercise caution in case any use or judgement is made of the model.

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# CONCLUSION

This project, *Student Academic Performance & Learning Risk Analysis*, explored how data mining and machine learning techniques can be used to identify patterns associated with students' academic risk from the dataset. Using the data set of 200 students, the Exploratory Data Analysis (EDA) was used to show some relationships between the students' academic risk and variables like attendance, sleeping time, internet time usage, and academics.

The K-Means clustering algorithm was applied to group the students into three clusters according to six numeric variables. The comparison of those clusters with the given AcademicRisk labels showed a high correlation with the high, medium, and low categories. The classification accuracy of the *Decision Tree classifier* is 77%. AssignmentScore and AttendancePercentage were the first two features used in its decision rules. Despite the fact that the model classified Low-risk students quite precisely, it was less successful at classifying Medium-risk students.

This project illustrated the necessity of such processes as *data preprocessing, features normalization, categorical encoding, model evaluation, and interpretability of prediction rules*. According to the documentation on scikit-learn library, K-Means makes clusters according to distances from data points, while Decision Trees give rules which can be used for understanding of the decision made [4, 5].

Nevertheless, the limitations of this research include the size of the database and features used. Results found by this research show connections but do not prove any causal relationships. Moreover, the results of models cannot be considered an assessment of individual students' risk. In the future, it is possible to conduct analysis using bigger database, finding more features and comparing other machine-learning algorithms.

In general, this project illustrated how exploratory data analysis, unsupervised clustering, and supervised classification may be combined for discovering patterns in educational data. With further testing and ethical application of such methods, they may help teachers to discover potential learning problems in students and opportunities to provide them with additional academic assistance.

# References

## Tools/software used:

- Link to the dataset: Copy of Student-Academic-Performance-Risk-Dataset.xlsx
- Orange Data Mining tool : Used for exploratory data analysis, K-Means clustering, Decision Tree classification, and model evaluation.
  Website: https://orangedatamining.com/
- WPS Office — WPS Spreadsheet — Used for viewing and checking the dataset and performing preliminary data-quality checks.
  Website: https://www.wps.com/


[1] Scikit-learn Developers. Preprocessing Data. Scikit-learn Documentation.
https://scikit-learn.org/stable/modules/preprocessing.html

[2] Scikit-learn Developers. Selecting the Number of Clusters with Silhouette Analysis on KMeans Clustering.
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[3] Scikit-learn Developers. Decision Trees. Scikit-learn Documentation.
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[4] Scikit-learn Developers. Clustering. Scikit-learn User Guide.
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[6] IBM. What Is a Decision Tree?
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[7] Google for Developers. Fairness: Identifying Bias.
https://developers.google.com/machine-learning/crash-course/fairness/identifying-bias


| Variable | High risk | Medium risk | Low risk |
|---|---|---|---|
| Study hours/week | 5.65 | 10.49 | 16.67 |
| Attendance (%) | 57.26 | 76.00 | 90.46 |
| Assignment score | 50.38 | 67.84 | 86.01 |
| Previous exam score | 45.21 | 65.93 | 81.65 |
| Sleep hours | 5.95 | 6.76 | 7.54 |
| Internet usage hours | 6.50 | 4.52 | 2.64 |